

10. In _____ queuing, packets are first assigned to a priority class. Each class has its own queue.
 A) FIFO B) weighted fair
 C) LIFO D) priority
11. _____ is the variation in delay for packets belonging to the same flow.
 A) Reliability B) Delay
 C) Jitter D) Bandwidth
12. The _____ bucket algorithm allows idle hosts to accumulate credit for the future in the form of tokens.
 A) leaky B) token
 C) either (a) or (b) D) neither (a) nor (b)
13. _____ is a class-based QoS model designed for IP.
 A) Integrated Services B) Differentiated Services
 C) Connectionless D) Connection-Oriented
14. In _____ queuing, the packets are assigned to different classes and admitted to different queues. The queues, however, are weighted based on the priority of the queues; higher priority means a higher weight. The system processes packets in each queue in a round-robin fashion with the number of packets selected from each queue based on the corresponding weight.
 A) FIFO B) priority
 C) weighted fair D) none of the above
15. In IPv4 header, an HLEN value of decimal 10 means _____.
 A) there are 10 bytes of options B) there are 40 bytes of options
 C) there are 10 bytes in the header D) there are 40 bytes in the header
16. An IPv4 datagram is fragmented into three smaller datagrams. Which of the following is true?
 A) The do not fragment bit is set to 1 for all three datagrams. B) The more fragment bit is set to 0 for all three datagrams.
 C) The identification field is the same for all three datagrams. D) The offset field is the same for all three datagrams.
17. UDP is called a _____ transport protocol.
 A) connectionless, reliable B) connectionless, unreliable
 C) connection-oriented, unreliable D) none of these
18. UDP and TCP are both _____ layer protocols.
 A) data link B) network
 C) physical D) transport
19. TCP groups a number of bytes together into a packet called a _____.
 A) user datagram B) segment
 C) datagram D) none of these
20. The connection establishment procedure in TCP is susceptible to a serious security problem called the _____ attack.
 A) ACK flooding B) FIN flooding
 C) URG Flooding D) SYN flooding

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

- | | Marks | BT* | CO* | PO* |
|---|-------|-----|-----|-----|
| 1. a) How is link cost set in link state routing? Explain with an example.
b) Imagine you have a group of mobile devices deployed in a disaster recovery scenario, where traditional communication infrastructure is unavailable. These devices need to communicate with each other to coordinate relief efforts. Illustrate this scenario with a suitable routing protocol that facilitates the establishment of routes between these devices when they are initially deployed and how it ensures the continuous maintenance of these routes as devices move around unpredictably in the dynamic environment. Illustrate. | 5 | L*2 | 1 | 1,2 |
| | 6 | L2 | 1 | 1,2 |

c)	Define traffic throttling. Explain hop-by-hop back pressure for congestion control.	5	L2	2	1,2
2. a)	Compare adaptive and non-adaptive routing in detail.	5	L3	1	1,2
b)	Explain anycast routing.	5	L2	1	1,2
c)	Explain admission control used in virtual-circuit networks to keep congestion at bay.	6	L2	2	1,2
3. a)	What is flooding in network routing? Explain how duplicate packets are controlled in flooding.	5	L2	1	1,2
b)	Mobile hosts, such as smartphones or tablets frequently move between different access points, like Wi-Fi hotspots or cellular towers, as they navigate the city. Illustrate this with a routing protocol to ensure efficient and reliable routing despite the dynamic nature of the hosts' movements.	6	L2	1	1,2
c)	Mention the drawback of utilizing choke packets for congestion notification, and then explain how ECN addresses and overcomes this limitation?	5	L2	2	1,2

Unit – II

4. a)	What is QoS. Explain how bandwidth, delay, jitter and loss requirements vary with the application types, with example.	5	L2	3	1,2
b)	List main features of IPv6. Draw the IPv6 header format.	6	L2	4	1,2
c)	List and explain ICMP message types.	5	L2	4	1,2
5. a)	Explain integrated quality of services.	5	L2	3	1,2
b)	List the characteristics of CIDR. Illustrate Route aggregation and longest prefix match concept in CIDR routing.	6	L2	4	1,2
c)	Explain Subnet Addressing. Find the SubnetID of the destination with IP address 150.100.15.11 and Subnet mask 255.255.255.128.	5	L2	4	1,2
6. a)	Explain fair queuing and weighted fair queueing.	5	L2	3	1,2
b)	Summarize six kinds of extension headers used in IPv6. How IPv6 supports jumbo payloads? Explain.	6	L2	4	1,2
c)	Explain Internet Multicasting.	5	L2	4	1,2

Unit – III

7. a)	Illustrate the relationship between the NSAPs, the TSAPs, and a transport connection.	8	L2	5	1,2
b)	With a neat state diagram explain simple connection management scheme in transport layer.	8	L2	5	1,2
8. a)	Explain service model of TCP.	8	L2	5	1,2
b)	Explain TCP congestion control mechanisms.	8	L2	5	1,2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
